

SETH FRIEND

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EDUCATION

University of Texas at Austin

Austin, TX

Electrical and Computer Engineering GPA: 3.79

Expected Graduation: May 2027

- Relevant Coursework: Digital Logic, Algorithms, Honors Linear Systems/Signals, Digital Signal Processing

EXPERIENCES

Hardware Engineering Intern

Phoenix, AZ

Honeywell Aerospace (ValFac)

June – August 2026

- Develop digital verification testbenches using SystemVerilog and UVM, driven by proprietary aerospace training
- Executed Acceptance Test Procedures (ATP) for a Bombardier hardware-in-the-loop simulation bench, supporting validation of real avionics hardware
- Engineered an automated Python data pipeline to parse SSI files and generate user-ready Excel workbooks, eliminating manual data formatting for ATP documentation

Electromechanical Lead

Austin, TX

Ivy Technology / Amazon Robotics

May – July 2025

- Led and coordinated teams of 7–10 to build 12 Cardinal automated work cells, delivering a week ahead of schedule
- Installed and wired 400+ automation-ready chutes in collaboration with other leads, integrating mechanical and electrical systems.
- Conducted quality assurance and troubleshooting on 20 Linux-based self-driving robots

Technical Assistant and Design Lead

Austin, TX

Texas Inventionworks | Design Team

Aug 2024 – Present

- Lead and support multidisciplinary student projects, providing guidance on embedded systems, fabrication, and design workflows
- Designed and led a CAD workshop and delivered hands-on training in soldering, 3D printing, and electronics reaching over 150+ students

Machine Learning Researcher and Robotics Mentor

Austin, TX

Robot Interactive Intelligence Lab

Jan – Dec 2024

- Achieved 95% success training a robot arm in simulation using imitation learning (robomimic)
- Collaborated on a team of 3 to integrate a motion capture system as an input device for Telemoma
- Configured Ubuntu/ROS environments for lab computers, supporting ongoing robotics projects
- Provided individualized help to 20+ high school students in PyTorch, ROS, and kinematics

PROJECT

Humanoid Robot | CAD, CAN Bus Communication, Control Systems, Simulation

July 2025 – Present

- Designed a humanoid robot in Onshape with 5 DoF in each limb
- Constructed robot using quasi-direct drive motors in each leg joint and servos in the arms
- Implemented CAN bus communication for synchronized motor control
- Developed a physics-based simulation environment in Isaac Lab to test reinforcement learning strategies
- Taught robot to stand while undergoing translational forces in simulation

EXTRACURRICULAR EXPERIENCES

Texas Robo Rumble | Team Lead (Annual Combat Robotics Competition Jan – Mar)

2024 – 2026

- Led a 7-member team designing and building 3 lb combat robots

Engineers for a Sustainable World | Ecobots Robotics Lead

Aug 2023 – Present

- Support new sustainable robotics project with arduino, robotic, and soldering workshops

Society of Hispanic Professional Engineers | Active Member

Aug 2023 – Present

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, Python, C++, C

CAD and EDA: Vivado, KiCad, LTspice, Onshape, Solidworks

Developer Tools and Libraries: Git, VS Code, Arduino IDE, Pandas, NumPy, Matplotlib, PyTorch